

Degree Project in Industrial Design Engineering

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Enhancing Reliability of Remote Fault Handling for Driverless Trucks

An interaction concept developed based on Systemic Design

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KTH Industriell teknik och management

Förbättring av tillförlitligheten hos fjärrsystem för felhantering av förarlösa lastbilar: Ett interaktionskoncept utvecklat baserat på systemisk design

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Sammanfattning

Idag spelar föraren en avgörande roll när lastbilar stöter på avvikelser. Det är de som upptäcker och agerar vid avvikelser, delar relevant information med intressenter och fattar beslut. Men vad händer när vi inför förarlösa lastbilar? Detta examensarbete undersöker hur ett fjärrstyrt felhanteringssystem kan utformas för att stödja drifttid och felhantering i förarlösa lastbilstransporter.

Systemiska designmetoder valdes för att utveckla förslag för ett så komplext produkt-tjänstesystem, eftersom de hjälper designers att integrera systemtänkande och användarcentrerad design. Genom att välja "Research through Design" som forskningsmetodik för projektets spekulativa område, utformades research processen för att iterativt och samarbetsinriktat utveckla designartefakter, som systemkartläggning, arbetsflöde och gränssnittsprototyper, för att visualisera en möjlig framtid och öppna upp för diskussion inom organisationen.

Genom litteraturstudier, fältbesök på Scanias servicecenter, semistrukturerade intervjuer med servicepersonal och två designworkshops med Scania-experter, kretsar detta arbete kring de utmaningar och möjligheter som finns med fjärrstyrd felhantering och AI-verktyg för beslutsfattande vid fordonsfel. Resultaten av projektet är: 1) ett systemdesignförslag för framtida felhanteringssystem för förarlösa lastbilar i ett TaaS (Transport as a Service)-sammanhang, där de viktigaste rollerna för mänskliga aktörer och intelligenta verktyg definieras; 2) designfaktorer för utveckling av digitala gränssnitt för felhantering, såsom att bygga transparens för att undvika "black-box"-effekter, spårbarhet för ansvar och etisk drift, skapa situationsmedvetenhet för team och individer, samt förbättra användbarhet och kopplingar mellan systemaktörer; 3) gränssnittsprototyper som demonstrerar dessa designfaktorer.

Detta examensarbete lyfter fram de sociala aspekterna av felhanteringssystemet och bygger vidare på kunskap om förtroendeskapande för felhantering i en förarlös kontext. Samtidigt visar det hur affärsvisionen kan påverka systemets design. För framtida forskning diskuterar författaren hur användar- och affärsperspektiv bör utforskas för att vidareutveckla förslaget.

Master of Science Thesis TRITA-ITM-EX 2024:542



KTH Industrial Engineering and Management

Enhancing Reliability of Remote Fault Handling for driverless trucks: An interaction concept developed based on Systemic Design

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Abstract

Today, the driver is essential when trucks experience a fault. They are the ones who detect and act when a fault occurs; they share relevant information with stakeholders and make decisions, for example. But what will happen when we introduce driverless trucks? This thesis project investigates how a remote fault-handling system could be designed to support uptime and fault-handling in driverless truck operations.

Systemic design methods are chosen to develop proposals for such a complex product-service system as they help designers integrate systems thinking and human-centered design. Choosing Research through Design as the research approach to explore this speculative field, the research process is designed to iteratively and collaboratively develop design artifacts, such as system mapping, workflow, and interface prototypes, that visualize the possible future and open up discussion within the organization.

Through literature reviews, field visits in the Scania service centers, semi-structured interviews with service center personnel, and two design workshops with Scania experts, this thesis builds on the challenges and possibilities of remote fault handling and AI tools for decision-making in vehicle fault situations. The results of the project are 1) a system design proposal for the future fault-handling system for driverless trucks in a TaaS (Transport as a Service) context, defining the key roles of human actors and intelligent tools inspired by the previous studies; 2) design considerations when developing the digital interfaces of fault handling, such as building transparency that avoid the black-box effect, tracing condition for accountability and ethical operation, providing situational awareness for team and individuals, increasing situational responsiveness and improving the usability of interfaces and connectivity between the system actors; 3) interface prototypes that demonstrate the design considerations.

This thesis highlights the social aspect of the fault-handling system and builds on the knowledge of trust-building for fault-handling in a driverless context. Meanwhile, it shows how business vision could impact the system's design. Suggesting future research, the author discussed how the user and business perspectives should be explored to develop the proposal further.

FOREWORD

This thesis is the final project for my Master of Science in Integrated Product Design at KTH Royal Institute of Technology, within the Industrial Design Engineering track. The project was conducted at Scania CV as part of the research project *System Transformation for Uptime for Driverless Vehicles (STIFF)*.

I would like to sincerely thank my supervisor, Lina Rylander, for her invaluable guidance, encouragement, and support throughout the fieldwork and workshops.

Thanks to all workshop and interview participants for contributing your time and insights.

Lastly, I want to express my heartfelt thanks to my family, friends, and my cat for their unwavering support during my studies. To my partner and best friend, Xucheng, who kept me grounded during the ups and downs, this journey would not have been as fun and enjoyable without you.

Jiepeng Wu

吴捷芃

Stockholm, October 2024

NOMENCLATURE

This chapter presents the abbreviations used in this thesis.

Abbreviations

IVHM	Integrated Vehicle Health Management
HCI	Human-computer Interaction
HMI	Human-machine Interface
TaaS	Transport as a Service
PSS	Product-service System
SMT	Service Management Tool
SA	Service Advisor
ST	Service Technician
CCO	Control Center Operator
DO	Diagnosis Operator
…	…

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1 INTRODUCTION

This chapter introduces the background motivation of this thesis project, along with the purpose and delimitation in order to outline the project scope and provide the research context.

1.1 Background and Motivation

Driverless vehicle is an innovation that can contribute to a safer and more efficient transportation system, which is transforming to be more environmentally, economically, and socially sustainable (Andersson & Ivehammar, 2019; Kristoffersson & Brenden, 2018). Researchers argue the need to adapt the service market with new prerequisites in a driverless context to enable safer and more efficient usage. One aspect that is important to achieve such sustainable transportation is reliability or uptime. Uptime is a measurement of the vehicle's functional state and the percentage of how much time the vehicle could be used and the time it was planned to be used (Bernspång & Kali, 2011). A fault-handling system supports uptime as a crucial vehicle maintenance and repair subsystem. The fault-handling system refers to actors and activities involved when a truck experiences a fault during an operational mission, including fault detection, detailed fault information collection, fault analysis, and decision-making on fault-handling actions (Rylander, 2023).

The driver is identified as a critical actor in the current fault-handling system when performing fault detection and diagnosis and plays an essential role in building the reliability of the fault-handling process (Rylander, 2023). How do we handle faults without the driver onboard in the future? This thesis focuses on developing an interaction concept for the fault-handling system for driverless trucks, based on the proposed system design concept from Rylander (2023), inspired by the Integrated Vehicle Health Management (IVHM) Framework. The IVHM framework, containing humans, products, services, technology, and cultural frames, aligns with the concept of a Product-Service System (PSS) defined by Morelli (2002). Designed for various purposes to fit the business model (Tukker, 2004), a PSS for vehicle health management aims to enhance profitability. Supporting vehicle uptime is crucial for operational profit, and an IVHM-focused PSS can improve maintenance and repair efficiency, thereby reducing costs and maintaining operation efficiency. IVHM as a framework inspired redesigning the fault handling system, which will be covered in Chapter 2.

In this transformation to a highly autonomous future, the challenge of defining the purpose of using technology is discussed (Dibsdale, 2011). One of the aspects is the role of Artificial Intelligence (AI). Industry 5.0 (European Commission. Directorate General for Research and Innovation., 2021) emphasizes integrating advanced technologies and human-centric design harmoniously. Industry 5.0 aims to enhance the collaborative interaction between humans and machines, focusing on the role of AI in creating symbiotic relationships where AI's analytical and processing capabilities augment human creativity and decision-making (European Commission. Directorate General for Research and Innovation., 2021). This synergy enhances productivity and efficiency and fosters an environment where human skills and well-being are

prioritized. It motivates the development of this project as it is more important than ever to identify the value and role of human wisdom in this evolving industry where AI and autonomous tools are being introduced into the workflow to support various tasks. This informs the design of the future handling system, emphasizing that AI should support the analysis and decision-making. At the same time, human expertise remains central to maintaining safety and efficiency and making final decisions regarding fault-handling action and vehicle health (Tao et al., 2024).

Redesigning the fault-handling system is crucial to supporting uptime in a driverless future. When key system actors change, other components, including processes and organizational structure, should adapt accordingly. In this thesis, I propose a fault-handling system adapted to the driverless environment and the case company's needs, identifying and trying to define the key actors—human and non-human—with certain delimitations. This thesis is more of an exploration of the possible future that helps spark discussions within the organization, contributing to the gradual transformation of future systems.

This thesis is part of the System Transformation for Uptime for Driverless Vehicles (STIFF) research project at Scania CV. Scania CV offers a wide range of commercial vehicles and services and is developing driverless trucks as part of the Traton Group. The STIFF project is a collaboration between academia and the industry to examine the requirements for Uptime for driverless vehicles.

1.2 Purpose

The study analyzes the current fault-handling system and conceptualizes a future system design based on findings in previous studies. Meanwhile, this study highlights the social aspect of the fault-handling system, emphasizing the importance of trust-building by defining the roles of intelligent tools and applying the proposed design considerations. The main research question is:

"How can we enhance the efficiency of fault handling and build trust when interacting in a remote setting for driverless trucks?"

This thesis is conducted with two objectives to answer the research question.

- The current system: To navigate the challenges and opportunities in this transformation regarding the relationship between humans, remote interactions, and intelligent tools by understanding the current fault-handling system.
- The future proposal: To propose and prototype a system design concept, developing knowledge on presenting information and supporting decision-making for remote fault handling.

1.3 Delimitation

This thesis is developed in a specific context. I focus on hub-to-hub transportation, which means that the operation task is performed between logistics hubs with highways and uses Scania's service networks. I focus on hub-to-hub transit due to the added complexity of handling faults remotely, especially when sending humans to the field, which is ineffective compared to other

types of applications, such as mining. This shows the value of exploring a remote fault-handling system that supports autonomous truck transportation.

Another context of this project is to focus on automation levels 4 and 5, which means that a driver no longer needs to be onboard to take over the driving, see Figure 1 (*SAE Levels of Driving Automation™ Refined for Clarity and International Audience*, 2021). Assumptions about the connectivity are made in which trucks are connected and perform remote communication and receive information regarding the vehicle, such as operational analysis, brake status, and error codes, which also has been suggested by Rylander (2023). This thesis project will explore the potential of remote detection and data collection. We assume that sensors for specific fault symptoms, such as sensors for colors of smoke, vibration, and sounds, could be placed as needed without interference from each other. Experts from the pre-study validated that it is possible to add sensors to the autonomous vehicle regarding accuracy and production budget.

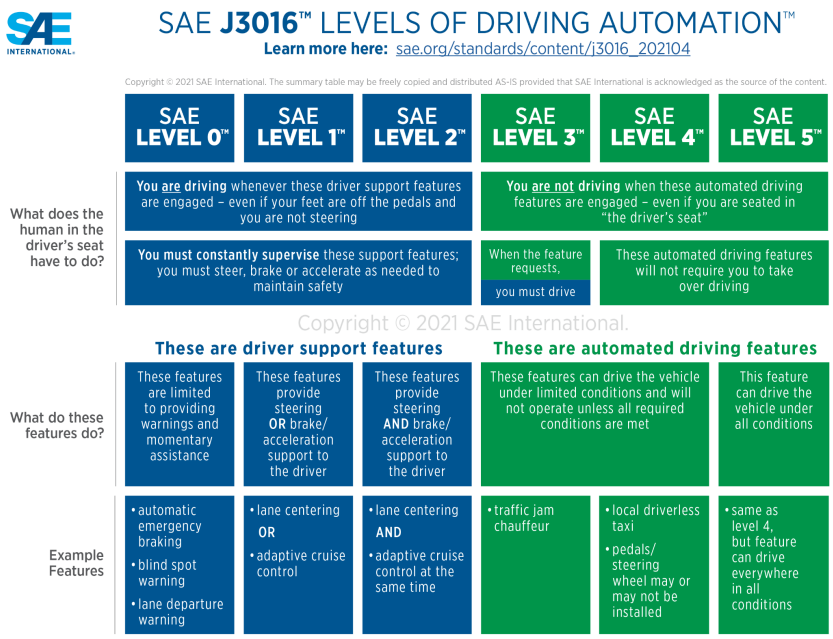


Figure 1. SAE Levels of Driver Automation (*SAE Levels of Driving Automation™ Refined for Clarity and International Audience*, 2021)

2 FRAME OF REFERENCE

This chapter establishes the theoretical foundation for this thesis project for understanding the current research and will be used to analyze the research findings.

2.1 Designing for Services

2.1.1 Transformation to a Service-Dominant Logic

Traditionally, the vehicle manufacturing industry has followed a Goods-Dominant (G-D) Logic, treating the vehicle as the primary output unit. In this case, services are provided around goods, and services were designed as if they were appendages to goods (Secomandi & Snelders, 2011). The current transport system can be considered a model with five layers: "material flow, transport operation, information operation, transport infrastructure, and telecommunication infrastructure"(Monios & Bergqvist, 2020, p. 11). In S-D Logic, the products are the carriers of the service, and the value is created through the service (Vargo, 2009). As vehicles are being applied with driverless technology, there is huge potential to create new prerequisites and expectations in their customer offerings. As Monios and Bergquist (2020) discussed, it will disrupt the current transportation system since the new transport system offers a network structure where information technology and intelligent software are the core of this new system. It drives the change in the business model into a Service-Dominant (S-D) Logic, which is more about doing things for and with each other (Morelli, 2002).

Transformation and transition often describe large-scale changes in a system. However, there is a difference between these two notions regarding the contexts of change (Hölscher et al., 2018). According to Hölscher et al. (2018), 'transition' describes sustainable changes brought by disruptive innovation, which drives an unsustainable condition into a sustainable one. 'Transformation' focuses on the societal change processes for the desirable change, which focuses on the individual values and motivation of the system actors. In this thesis, 'transformation' is used to describe the changing processes of the fault-handling system for the future of autonomous transportation.

2.1.2 A result-oriented Product-Service System

Product-Service Systems (PSS) represent a paradigm shift from the traditional Goods-Dominant (G-D) logic, primarily focusing on tangible and intangible products (Baines et al., 2007). Many view Product-Service Systems (PSSs) as an excellent means to simultaneously enhance competitiveness and promote sustainability (Tukker, 2004). PSS combines products and services to jointly fulfill user needs, offering a more integrated solution (Morelli, 2002). This concept is built from various perspectives, including marketing, service management, and product management. From a marketing perspective, PSS addresses the transformation from focusing on material products to a blend of products and services. In product management, PSS involves either extending the service component around traditionally product-oriented business

activities or introducing a new service component in service-oriented businesses. Baines et al. (2007, p. 1545) illustrate the key aspects of PSS by referencing Goedkoop et al. (1999):

- "1. Product: a tangible commodity manufactured to be sold.".
- "2. Service: an activity (work) done for others with an economic value and often done on a commercial basis.".
- "3. System: a collection of elements including their relations.".

According to Monios and Bergqvist (2020), the vehicle manufacturing industry is shifting from the current transportation model toward PSS. This model sells transport as a service instead of merely selling trucks. In this future system, the vehicles will serve as the product enabling the services, but the freight transportation process will be marketed and sold to customers as a service. To further explore what end value can be conveyed to consumers through PSS from a business model, I referred to Tukker's (2004) classification of PSS systems, which vary on a spectrum where, on one end, the main value rests on product content (tangible) and, on the other, on service content (intangible).

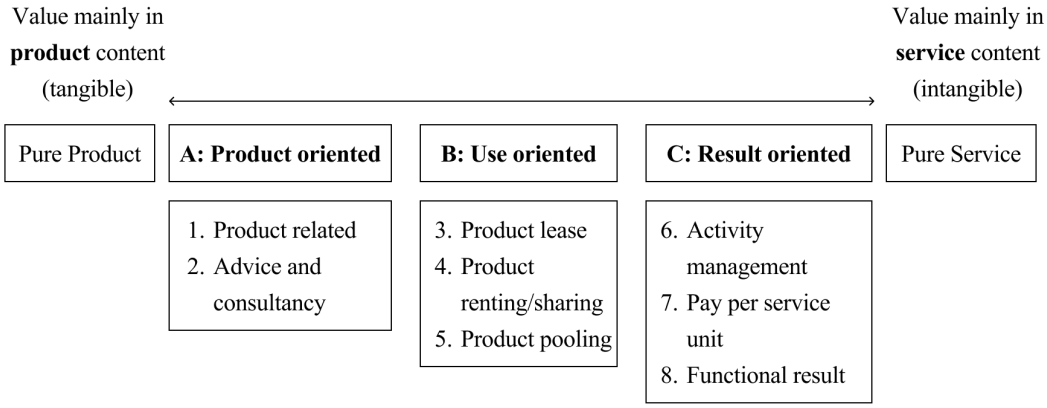


Figure 2. Main and subcategories of PSS (Tukker, 2004)

There are three main categories of PSS within the spectrum (Tukker, 2004, p.248):

- Product-oriented: The business model is still mainly geared towards product sales, but some extra services have been added.
- Use-oriented: The traditional product still plays a central role, but the business model is not geared toward selling products. The product stays in ownership with the provider, is made available in a different form, and is sometimes shared by several users.
- Result-oriented: The client and provider, in principle, agree on a result, and no predetermined product is involved.

The current business model in road freight, in which the manufacturers mainly sell vehicles to transport providers and provide after-sale services such as fault-handling service to optimize product usage (Monios & Bergqvist, 2020), could be categorized as Product-oriented with a primary goal of selling products and services are added as 'extra' (Tukker, 2004). Studies regarding the shift from a product-oriented to a service-oriented ecosystem have been conducted in the field of autonomous transportation, which refers to the business model where customers can access vehicles on demand without needing ownership (Duarte et al., 2024). Such a shift emphasizes convenience and flexibility, allowing users to utilize vehicles only

when needed, thus reducing the overall cost and environmental impact associated with car ownership (Duarte et al., 2024). Transport as a Service (TaaS), proposed by Traton, addresses a broader, subscription-based approach that includes a full suite of services beyond vehicle access. This model covers not only the vehicle itself but also includes insurance, maintenance, and charging infrastructure (*TRATON – Transport-as-a-Service*, 2024). In this case, TaaS could be regarded as a user-oriented model since the product still plays a central role. Still, the business model is sorely selling a product, but the service package surrounds the products. This approach reflects a shift towards a more inclusive and streamlined service experience, meanwhile pushing the changes of service scope of fault-handling to be more extensive, compared with the current business model in which the manufacturers are only responsible for a part of the lifecycle of the vehicles.

2.1.4 Identify the social structure in the Fault-handling system

Systemic design is an initiative in design that integrates systems thinking and human-centered design; it helps designers to scope the project and describe complex and multi-stakeholder systems such as transportation, medicine, and healthcare, addressing the complex and strained social systems (Jones, 2014; Vink, Wetter-Edman, et al., 2021). It brings an action-oriented approach to change, emphasizing a plurality of methods rather than a singular best approach. This resurgence of systems thinking under the systemic design banner focuses on mapping the complexity of systems through tools like giga-maps and synthesis maps, which identify actors, materials, information, roles, and relationships within a system (Sevaldson, 2022).

In 2002, Morelli discussed critical characteristics of PSS, emphasizing the collaborative relationships between users, designers, and service providers, the synchronous nature of production and consumption in services, and the intangible nature of services compared to the tangible nature of products. These characteristics highlight important societal aspects, in this case, considering functionalities of the fault-handling system such as fault detection, information exchange, and decision-making (Rylander, 2023). Furthermore, Vink et al. (2021) argued that strengthening and tapping into social structure is a critical leverage point for systems change and why understanding and intentionally shaping social structures are critical for influencing social systems.

Social structure has been discussed from a systemic design perspective (Vink, Wetter-Edman, et al., 2021) and organization theory (Derbali, 2014). Derbali (2014) discusses social structure as a set of non-fortuitous social relations between individuals, parties, and an organization. Business practices can be understood as social structures, which are defined as organized sets of non-random social relations among individuals and groups within an organization. Power relations and social reporting in business social structures involve a hierarchy of authority, division of labor, and formal rules and procedures. Wilmering and Davies (2011) also discussed the ‘law of unintended consequences,’ which states that the new vehicle management platform could change behavior in business scenarios. By understanding why different roles in the fault-handling system behave the way they do, in other words, how these social structures work, we could identify the critical point of intervention that drives and resists change. Vink et al. (2017) discussed the role of system actor’s inner representations, known as mental models, arguing that mental models play a critical role in enabling system behavior and transformation. Using design objects to engage actors and trigger related emotions can allow actors to open up discussion and rethink the system structure (Vink et al., 2017).

Seeing the fault-handling system as a social structure reveals the emotional aspect, especially trust, which researchers have discussed (Dibsdale, 2011; Rylander & Englund, 2024; Tao et al., 2024). The relationship between the emotions of human actors and Intelligent tools is an inevitable topic when applying intelligent tools where humans are needed in the workflow. For example, for remote data reading, some respondents found it difficult to trust the data without access to the driver, as the driver is the 'eyes and ears' (Englund, 2021). While seeing the absence of the driver as the challenge, there are opportunities in designing a trust-worthy product-service system for autonomous fields (Calvert et al., 2023; Dibsdale, 2011; Englund, 2021; Rylander & Englund, 2024). Calvert et al. 2023., for instance, present a framework for achieving 'Meaningful Human Control' (MHC), which underscores the importance of human oversight in autonomous systems. Englund (2021) and Xin et al. (2024) argued that there is potential to earn trust through technology to remedy the lack of fault detection due to the driver's absence.

2.2 A Fault-handling System Based on IVHM Framework

2.2.1 Integrated Vehicle Health Management

This thesis suggests using the Integrated Vehicle Health Management (IVHM) framework to develop design considerations for the future fault-handling system. The definition of IVHM is (Jennions, 2011, p. 3):

> *"The unified capability of a system of systems to assess the current or future status of the member system health and integrate that picture of system health within a framework of available resources and operational demand."*

The author further examines the role of technology in business models, emphasizing that the benefits of technology must be actively managed. This often requires changes within the organization and the supply network. Jennions (2011) discussed that, in an IVHM Framework, the social aspects must be understood to determine how technology can support the business model.

2.2.2 Two main actors in fault handling system

The value of IVHM technology lies in its ability to collect information on an object's condition, performance, and location and in facilitating the transmission of this information (Jennions, 2011, p. 4). A control center has been suggested as a service provider for remote assistance for driverless vehicles through organizing and executing remedial actions (Dibsdale, 2011). The functionalities of a control center could be monitoring vehicle health, isolating the cause of issues, identifying the location of faults (such as specific components or systems), and determining appropriate corrective actions (Augustin, 2013; Rylander, 2023).

Inspired by the IVHM Framework (Jennions, 2011), researchers proposed a fault-handling system design concept with two main actors in a control center for driverless trucks: a *Control Center Operator* and a *remote Diagnosis Operator* (Englund, 2021; Rylander, 2023; Rylander & Englund, 2024). The responsibilities for the two roles are inspired by the typical organization of haulage companies, which usually have a traffic manager and a vehicle manager. The Control Center Operator handles mission planning and is the primary decision-maker. A Diagnosis

Operator, who is more of an advisory role when trucks experience a fault, will support the Control Center Operator with a remote diagnosis system (Rylander, 2023).

Dibsdale (2011) proposes a resource optimization strategy that involves centralizing a multidisciplinary team to manage vehicle fleets. This approach enhances mutual understanding of each team member's tasks and reduces unnecessary back-and-forth communication, allowing for the early correction of mistakes or misunderstandings. This concept inspired the room setting and offers benefits such as cost reduction, increased revenue, and lower risk.

2.2.3 Data quality

Data quality is one of the important aspects when developing the tools for the Control Center, and it represents the way to present data that fit for purpose in a particular context subjectively (Dibsdale, 2011). To enable shared situational awareness, an interface displaying critical data should facilitate rapid appraisal and readiness for any unit member. This interface should provide situational awareness. Situation awareness was originally applied to military pilots whose lives and missions required an especially high awareness of the details around them (Few, 2013, p. 29). It means being aware of what is happening around the user and understanding what that information means now and in the future (Endsley, 2011). It is suggested that a fleet view, showing the condition of all vehicles, could be presented to convey a sense of "being in control" (Dibsdale, 2011, p. 116) with “status at a glance” when the humans have been “out of the loop” (Calvert et al., 2023, p. 8). Dibsdale (2011) proposes design considerations regarding the diversity of user profiles, user roles from the business perspective, usage patterns, and data type, which will be discussed in Chapter 4.

Prototyping the interfaces, in this case, referring to the information dashboard, is one of the objectives of this thesis. According to Few (2013), an information dashboard is defined as

"a visual display of the most important information needed to achieve one or more objectives that have been consolidated on a single computer screen so it can be monitored at a glance (Few, 2013, p. 26)"

As proposed by Few (2013), situation awareness can be supported on three levels: Level 1—perception of environmental elements; Level 2—comprehension of the current situation; and Level 3—projection of future status. This multi-level approach ensures that dashboards display data effectively and support comprehensive situational awareness and informed decision-making. These frameworks are applied when designing the wireframe and user flow of the prototypes, with a design goal of updating high-level situational awareness first, then identifying and focusing on specific items requiring attention.

2.2.4 Intelligent tool and human decision-making

The application of intelligent tools is a big part of technology that supports information analysis and decision-making. One challenge here is to define the roles of human actors and intelligent tools in decision-making. Once the industry has unfolded the power of intelligent tools and begun utilizing their capabilities, it becomes particularly important to explore human actors' advantages and evolving roles in the workflow. More specifically, it is crucial to understand how humans and AI can better enhance each other's complementary strengths and propensity to ‘act synergistically’ and avoid the ‘black box’ of both humans and algorithms decision-making (Bonezzi et al., 2022; Jarrahi, 2018, p. 577; Koutsikouri et al., 2023). Koutsikouri et al.

discussed the values of human judgment in decision-making, see Table 1. Among these, not-knowing, lived experience, and episteme are valuable in this thesis, especially in the transformation where new vehicles, new sensors, and other technologies are applied in the service delivery process. However, human decision-making risks projecting personal intuitive understanding and causing the 'black-box' effect (Bonezzi et al., 2022). Here, we see the potential of using Intelligent tools with a sense of their limitations to improve the efficiency and quality of fault-handling while avoiding the "black box" effect in human decision-making when prioritizing urgency level, analyzing faults, and making decisions.

Table 1. Seven Elements of judgment in human and autonomous Decision-making (Koutsikouri et al., 2023)

Elements of judgement	Human decision-making	Automated Decision-making
Not-knowing	Relating to and being aware of not knowing all aspects of a situation.	Uses only known data. Unable to change ultimate goals.
Episteme	Theoretical knowledge that is used in a particular setting.	Knowledge that is represented in the data can be used for a specific purpose
Lived experience	Using experiences (self and others) to mitigate against biases and prejudice.	Experiences based on already processed data, with or without biases. Ability to 'reflect' on past experiences based solely on previous data usage.
Intuition	Using different types of sensory data and previous experience, we can identify patterns that are essential for a particular situation, even if they cannot be fully articulated.	Capacity to process immense amounts of data to see patterns
Sensory perception	Remaining open to the incoming sensations despite the influence of prior knowledge and prejudice.	Disconnected from lived sense-making and cannot create meaningful connections
Emotions	It informs what is essential in a situation and motivates what to do next.	It does not possess emotions and hence does not have the capacity for emotional involvement required for human-centered decisions.

Techne	Using prior lived experiences that are systematized.	Using manualized, standardized, and/or formalized knowledge.

3 METHODOLOGY

This chapter starts by presenting the theoretical approach and principles behind the research design, followed by the procedures and methods used in this study.

3.1 Research Methodology

This thesis aims to propose a fault-handling system for the future through understanding and interpreting the experience in the current fault-handling system, discovering design considerations, and collecting inspiration from different perspectives in the organization. Qualitative research suits this purpose since it is associated with discovery, understanding, and shared interpretation (Frankel & Racine, 2010; Sanghera, 2007). Furthermore, qualitative research methods are used to explore and understand human behavior and the reasons behind the behavior (Creswell, 2009; Frankel & Racine, 2010).

This thesis explores how to handle faults to support uptime in a driverless future and opens up ideas on how the desirable near future could potentially be. Research through Design (RtD), regarded as a way to explore the speculative future, has been chosen as the approach to explore this speculative field with the idea of using design's strength as a reflective practice of continually reinterpreting and reframing a situation to investigate the future (Zimmerman & Forlizzi, 2014). Here, speculative is a term used to represent "a direct correlation between 'here and now'" (Auger, 2013, p. 12), which is what this thesis project looks for - developing the future-oriented proposal based on understanding the current system.

RtD looks like design practices on the surface because it performs a process of making and critiquing artifacts that function as proposed solutions (Zimmerman & Forlizzi, 2014). However, instead of using theory to drive the creation, RtD uses design artifacts to proceed with theory and generate new knowledge. Design artifacts, implied as conceptualization in design practices, could be regarded as storyboards, prototypes, maps, and trigger materials (Savic & Huang, 2014). Meanwhile, Vink et al. (2017) argued that using heterogeneous objects and designing objects in this context can engage actors' senses and open discussion. This project is considered a reflective process to develop design artifacts that combined with explanations describing the system functions and modes of interaction rather than proposing an optimal solution or a definitive outline of capabilities (Auger, 2013; Mitrović et al., 2021; Zimmerman & Forlizzi, 2014).

As Vink, Koskela-Huotari, et al. (2021) and Sevaldson (2022) highlighted, systems are not static; they continuously adapt and change as actors interact within them, shaping and being shaped by institutional structures. With the nature of shaping institutionalized social structures to facilitate desired change, Service Design is chosen in this project as the method to investigate the complexity and changes and conceptualize the service system, providing insights on how the actors co-create value within the service system to drive intentional change (Vink, Koskela-Huotari, et al., 2021; Vink & Koskela-Huotari, 2022).

With the lens of Systemic Design, the fault-handling system can be visualized as an overall shape instead of the sum of its fragments (Sevaldson, 2022). Mapping out the main actors, the structure and shape of the service, rather than all the elements in the services, encourages the view of the system as design artifacts, which could be used thoroughly in this reflective process (Sevaldson, 2022; Zimmerman & Forlizzi, 2014).

To understand mental models that shape the current system and enable transformation, this project used design objects to trigger discussion and rethink the system structure, aligned with the RtD approach (Savic & Huang, 2014; Vink et al., 2017). Design artifacts are used as tools to trigger discussion on the existing system structures, fostering a deeper understanding of the potential for transformation and improvement within the service system. For example, presenting the storyboards of the imaginary future of fault handling for driverless vehicles triggers the interviewees to compare their current experience with the imaginary ones and to discuss their expectations, attitudes, and trust level of this transformation.

Collaborative research is applied to develop a solution with the researcher (Rylander, 2023). In this project, design methods were chosen to involve different perspectives when developing the design objects. People working on related products and services are involved, such as Service Advisors, Technicians, designers, product owners, and engineers of digital products and services.

3.2 Research Method

As mentioned in Chapter 1, this study is part of a research project at Scania CV. The expert interviews, fieldwork, and workshops were all conducted within Scania CV.

The research process was designed inspired by Research through Design. It aimed to use design artifacts, in this case, the visualized concept and prototypes, to explore solutions for future fault handling. The fieldwork and workshops were conducted to develop the artifacts further, as illustrated in Figure 4.

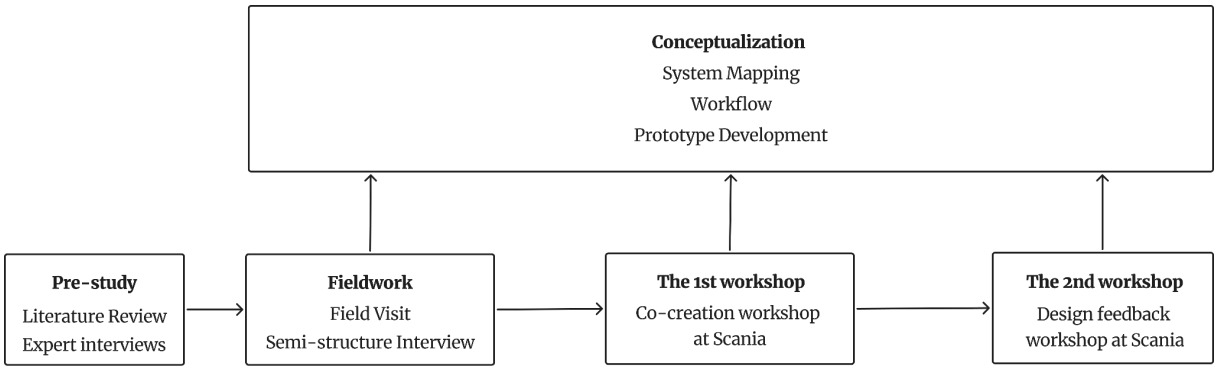


Figure 4. Research process

3.2.1 Pre-study

This project started with literature studies of papers from credible databases like Web of Science and Scopus to understand the research context. The references used in this project are research

articles from scientific journals and academic conferences. Keywords such as “Systemic Design, Human-Machine Interface (HMI), Service Design, Research through Design, Double-diamond model, Embodied knowledge, Mental Model, Trust, Speculative Design, Transport as a Service, and Autonomous vehicle” were used for the search.

Meanwhile, at this exploratory stage of the thesis, expert interviews were conducted within Scania CV with people who work on mechanical engineering, human factor research, and data science of autonomous systems to understand the constraints and delimitation of this thesis and how the outcome could benefit the future development of the future fault-handling system.

During the expert interviews, one of the remaining questions is defining the user in the future fault-handling system, specifically the user needs and competencies for this unexciting role in the current system. In the current system, two main roles, Service Advisor (SA) and Service Technician, are identified as potential user groups, which inspired the sample design and interview guide design for the semi-structured interviews. The current system mapping and workflow were developed to present the interactions and relationships between these roles.

3.2.2 Fieldwork

Field visits and semi-structured interviews were conducted to understand the current experience of fault handling and identify the opportunities and challenges when developing the future system. Field visits were conducted at two Scania service centers in Stockholm and Enköping, and five semi-structured interviews were conducted with Service Advisors and Service Technicians. The research design for the interviews is presented in Table 2.

Informed by the pre-study, an interview guide for each potential user group was developed and aimed at the following purposes:

- Validate the fault handling system workflow, focusing on the interaction between the driver, Service Advisor, and Service Technician.
- Understand the attitude and behavior of digital tools, like the Service Management Tool (SMT), and identify the challenges and opportunities for fault-handling systems in a driverless context.
- Explore the possibility of involving potential users in the participatory design process.

Table 2. Research Design of Semi-structured Interview

Role	Questions	Sample Size
Service Advisor (SA)	Understand how the user feels about the **remote** diagnosis, what information is missing in the current system, and what extra information they need from the drivers.	N=3
Service Technician (ST)	What information could only be collected when the vehicle is in the workshop? What information could only be collected when you try in the car like a driver?	N=2

During the pre-study, two roles in the current system were considered crucial to fault handling, and the interview guides were designed with different focuses. The interviews with Service Advisors focused on understanding their experience with remote data reading and their attitude toward the driverless future. During the first interview, the respondents were asked to go through the dashboard information session from the Service Management Tool so that I could understand what information is presented remotely nowadays, which helped to make the following interviews more efficient. The interviews with Service Technicians focused on understanding how they trust the information from drivers and remote data reading and exploring the possible ways to present the inexplicit information as they have more technical knowledge and are more confident about the symptoms of the trucks. The interview guide can be found in the Appendix.

Trigger materials and storyboards were used during the interview to stimulate the interviewees' discussions of perspectives and facilitate efficient communication, as presented in Figures 5-7.

- Images of autonomous vehicles were used to start talking about the vision and attitude of the driverless future.
- A list of symptoms that nowadays cannot be read remotely was presented to highlight the most common and crucial ones for making the fault handling decision.
- Two fault scenarios were presented to talk about their workflow nowadays.
- Two future scenarios generated by Copilot were used to trigger their imagination of fault-handling in the future.



Figure 5. Image of a truck used nowadays (left) and a conceptualized autonomous truck (right)

What are the most common and important symptoms according to your experience of trouble shooting? Any relevancy between them?

1. Corrosion
2. Vibration
3. Visual fault
4. Seizing
5. Noise
6. Color of smokes (white, blue, black)
7. Power issue, running uneven
8. Control non-functional
9. Paint damage

Fault Case 1 Engaged Brakes

• BASIC SYMPTOM: The vehicle pulls sideways
• ADDITIONAL SYMPTOMS: The vehicle runs sluggishly, marks from the wheel on the ground, the brake disc is hot

Scenario Description
• The driver has been driving a truck for many years and usually drives the same truck.
• A working day, on the motorway between Södertälje and Jönköping, the driver **feels that the vehicle pulls to the left**...and slows down.
• The driver instinctively/casually checks the rear-view mirror and **sees streaks on the ground** behind.
• The driver realizes that something is not right.
• He stops at the next rest stop...and calls the workshop and informs client about the symptoms he experienced.
• The customer receiver asks the driver to **check the tires** and the like to **see if it is possible be punctured**, something has become **tangled** or if a **wheel is hot**.
• The driver gets out and notices that the brake disc on the right is very hot front tire.
• The customer receiver concludes that there may be a fault with the brake caliper or similar that causes the brakes to apply.
• The driver is instructed to wait for recovery.

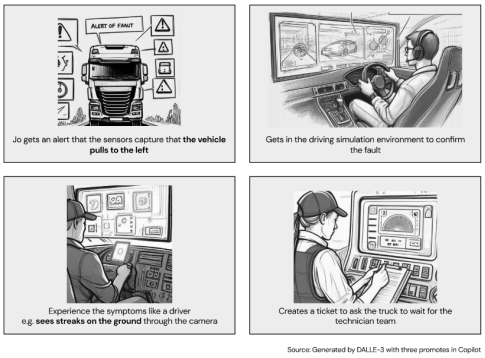
Fault Case 2 Air Leakage

• BASIC SYMPTOM: Warning light shows "High air consumption"
• ADDITIONAL SYMPTOMS: Squealing noise, long start-up

Scenario Description
• The driver has been driving a truck for many years and usually drives the same truck.
• When the driver has to start his route after the car has been **loaded, it takes a little longer** than that usual to start the vehicle, but it is not something that the driver thinks about anymore.
• Later in the day, on the highway between Södertälje and Jönköping, see the driver a **white warning of high air consumption** and knows that this is usually not something to care about.
• Further a little later, when the driver unloads goods at a transshipment center, so the driver hears a **hissing noise** from the vehicle.
• The driver therefore decides to consult the customer receiver at the workshop about what has happened.
• The driver is then recommended to go to the nearest workshop.

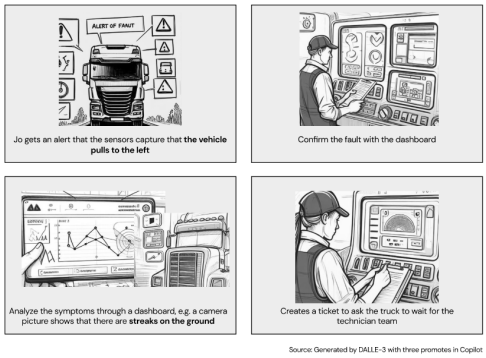
Figure 6. Symptom list and fault scenarios from Scania that were used during interviews

CONCEPT A: YOU ARE THE 'DRIVER'



Source: Generated by DALLE-3 with three promotes in Copilot

CONCEPT B: DASHBOARD



Source: Generated by DALLE-3 with three promotes in Copilot

Figure 7. Storyboards presented future scenarios

Two researchers presented during the field visits and interviews. One was the moderator who led the interviews. The other was the contact with service centers who coordinated time and supervised the interviews. Notes were taken and later organized in a table with themes. The findings were summarized as themes and implications for the next step.

3.2.3 The first workshop - Co-Creation Workshop at Scania

Two workshops were conducted throughout the project. The first one, a co-creation workshop, was conducted to invite different perspectives to define the capability and user of the future fault-handling system and to identify the stakeholders and competencies for the next workshop.

The workshop was conducted at the Scania office in two hours. Five respondents from Scania were invited from various areas, including people designing Service Management Tools for Service Advisors, autonomous systems for other applications, and remote control experience; see Table 3. During these two hours, the insights from fieldwork and pre-study were presented, and the initial system design was proposed and discussed.

Table 3. Workshop participants

Respondent	Role at Scania	Responsible product platform
1	Software engineer	Troubleshooting Tool for Service Technicians
2	Software engineer	Control Center development for Autonomous Truck
3	UX Design	Control Center development for Autonomous Truck
4	Product owner	Service Management Tool (SMT) for Service Advisor
5	Product owner	Remote control platform

A PowerPoint presentation, printed materials, and moderator scripts were prepared. The workshop was conducted with the following steps:

- Firstly, I introduced the project background and objectives of this workshop to provide the context for the participants. I then presented the findings from literature reviews, previous studies, and field visits to feed in more information. The current system was visualized with a focus on three key roles: the truck with the fault (driver), fault handling role (Service Advisor), and technical support (Service Technician) (see Figure 8).
- Secondly, a system design proposal developed based on Rylander's (2023) proposal was presented in a simplified and visualized way, using the same visual style and color representation (see Figure 8). We discussed the system design regarding who should be involved and how each actor should interact with each other.
- After discussing the systemic perspective, we zoomed into the fault handling role, seen as the Green part in Figure 8. We provided a tool to define a user statement (*User Need*

Statements, 2019) with three triggering questions (see Figure 9). The respondents wrote their ideas on post-in individually and put them on the whiteboard to present and discuss with each other.

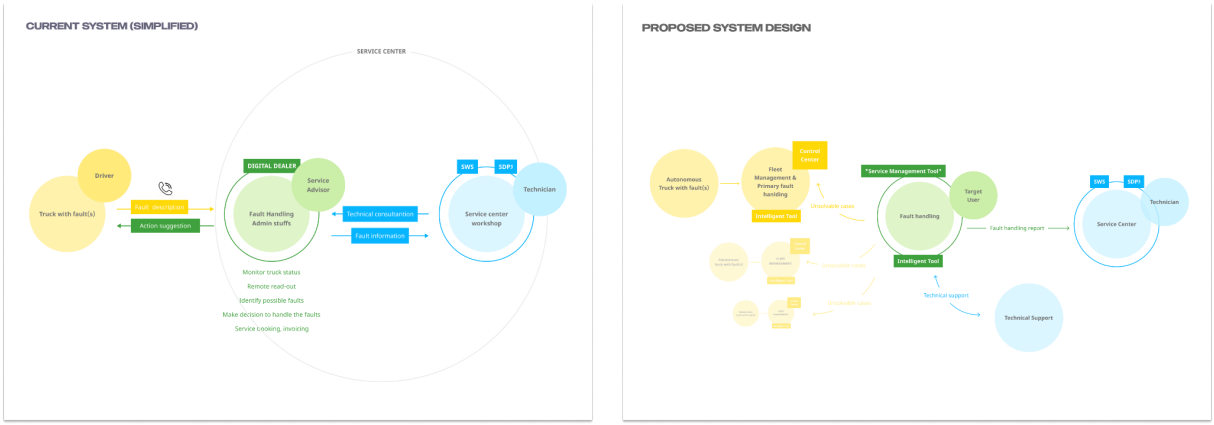


Figure 8. The current system mapping (left) and proposed system mappings (right)

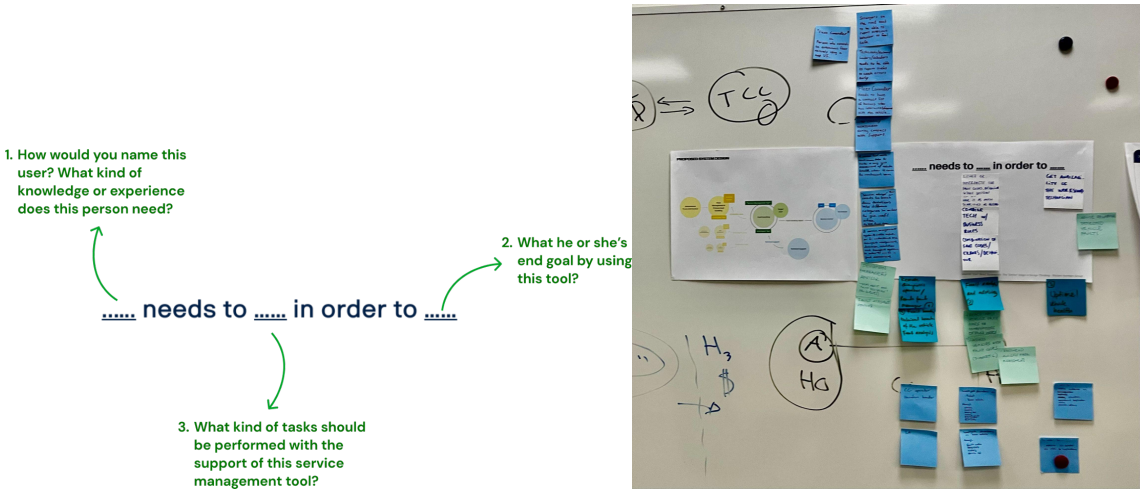


Figure 9. Triggering questions used to define user statements and ideas from the respondents

3.2.4 Conceptualization

The purpose at this stage was to develop design artifacts to demonstrate what a near-future fault-handling system may look like based on the findings from the previous research activities. As mentioned in Session 3.1, such conceptualization is not necessarily part of product development for launching; it serves as a tool that visualizes the findings to provoke conversation during the workshops and future studies (Lurås, 2016; Zimmerman & Forlizzi, 2014). There are three parts of conceptualization:

- A refined system map developed further from Figure 8.
- Inspired by the Business Process Model and Notation (BPMN), the workflow for handling a fault or deviation alert was developed to propose the fault-handling process when the service provider takes care of the vehicle throughout the product's lifetime which refers to Transport as a Service (TaaS).

- Two interfaces inspired by previous studies are proposed to support the proposed workflow. Figma is the primary tool for prototyping the wireframe and UI mockup. The graphical representations of system details and AI diagnosis are derived from the work of Rylander and Englund (2024) and Xin (2024).

3.2.5 The second workshop - Design Feedback Workshop at Scania

This project's second workshop was conducted to gather feedback on the design artifacts, in this case, the system map, workflow, and prototypes. Participants were from different departments, as described in Table 4; most were present at the first workshop, with one more Service Designer (Respondent 4). Since this workshop aimed to provide feedback on the artifacts developed further from the previous stage, it was beneficial to invite some of the people from the first workshop to facilitate the discussion efficiently without introducing the background and insights again. The roles of participants are listed in Table 4.

The expected outcome of this workshop was to collect feedback from usability, technical, and business perspectives so that the prototype is relevant to the real-case scenario and aligns with the design considerations. Based on the feedback, several changes to the workflow and prototypes were made, and the revised version is proposed as the final design of this project. A presentation was made with Figma Slides; see Figure 10. Feedback was summarized and used to refine the design proposal.

Table 4. Workshop participants

Respondent	Role at Scania	Responsible product platform
1	Software engineer	Control Center development for Autonomous Truck
2	UX Design	Control Center development for Autonomous Truck
3	Product owner	Service Management Tool for Service Advisor
4	Service Designer	Logistic related platform

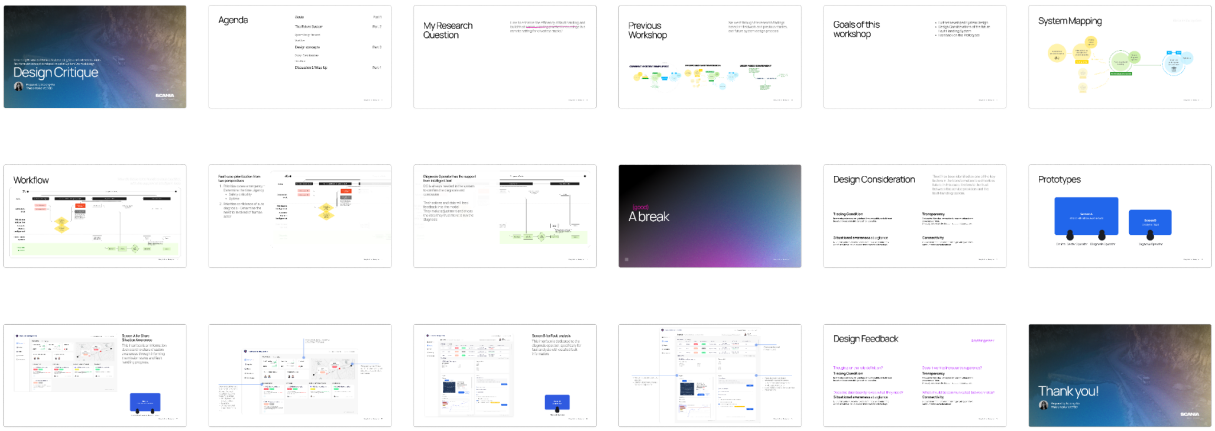


Figure 10: Screenshot of slides used in the Design Feedback Workshop

4 RESULTS

This chapter presents the results, starting with an analysis of the current system, followed by the proposed future system mapping, the fault-handling workflow, and the prototypes of interfaces that support the core fault-handling process.

4.1 Understanding of the Current System

4.1.1 The current system

The current scope of fault handling is based on a Goods-Dominant (G-D) Logic, where the fault-handling service is provided around the vehicles. The fault-handling scope only covers the time the trucks are on a mission. Drivers have been identified as crucial in the current fault-handling system (Rylander, 2023). Meanwhile, there are two main roles in the Scania service center: the Service Advisor (SA) and the Service Technician (ST). In the current fault handling system, Service Advisors, who are in direct contact with the driver, collect information remotely from drivers and Service Management Tool (SMT), analyze the fault, and provide advice on how to react to the fault. With more knowledge of the vehicle faults, ST takes care of the fault for further diagnosis and repair without direct contact with the drivers. Table 5 presents their responsibility in the current fault-handling system and the digital tools they use.

These two roles work together to handle a fault during the mission with the driver's involvement. The system mapping, as Figure 11, illustrates the relationship between the driver, SA, and ST. When a fault symptom appears, the driver contacts the SA for suggestions for handling the fault if the driver can’t solve it. SA collects the fault information through phone calls with the driver and remote data from SMT. The SA will then analyze the fault and suggest how to handle it. Sometimes, SA asks ST for suggestions on the fault analysis. Depending on the severity of the fault, the truck will be sent to the workshop if there is a risk of a major breakdown. ST will take care of the repair. This workflow is illustrated in Figure 12.

Table 5. Responsibility and tools

Roles	Responsibility	Tool
Service Advisor (SA)	They are in direct contact with the driver, who collects the information from human actors, gives initial advice, and collects the information from drivers for service personnel.	A Service Management Tool (SMT) performs remote reading and reads the truck's information.
Service Technician (ST)	There is no direct contact with the driver, but he needs the information to perform a diagnosis. Interact with the Service Advisor, who takes care of the error, repair, and maintenance.	ST uses a troubleshooting tool to read the truck data while repairing and maintaining it.

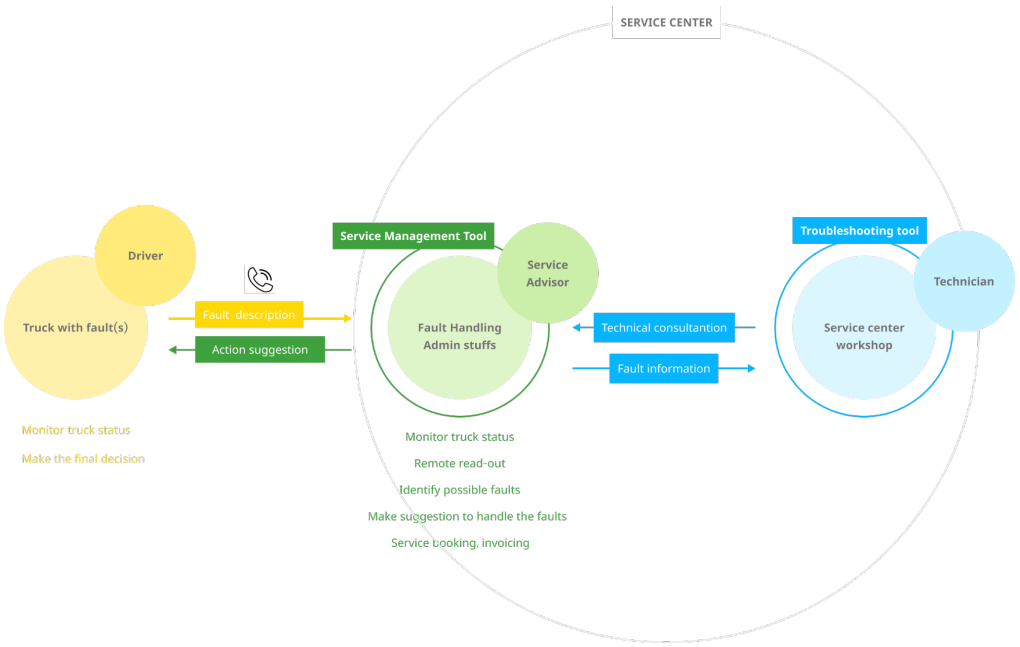


Figure 11. System mapping of the current fault-handling system

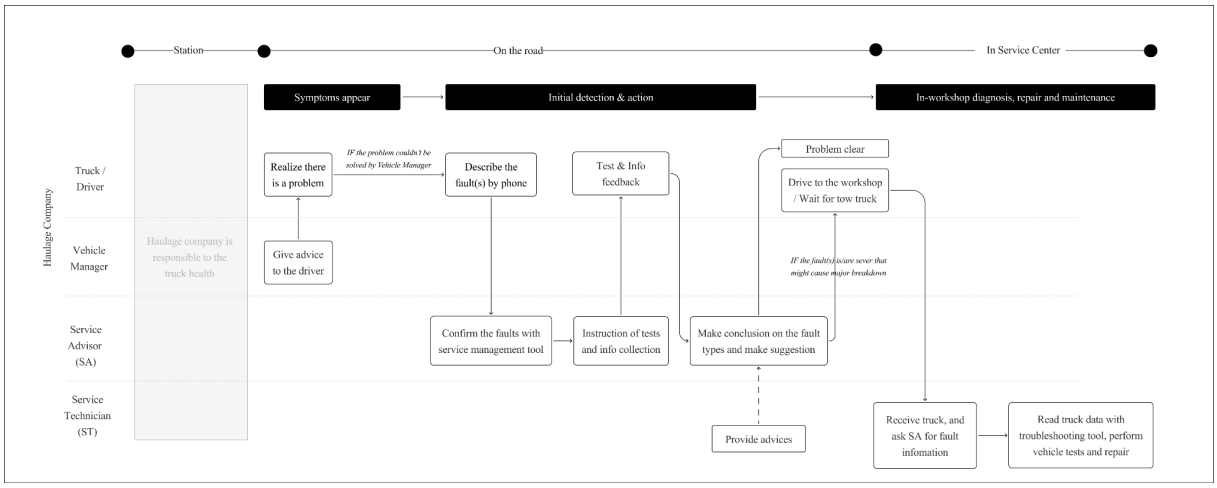


Figure 12. The current fault-handling system workflow

From the Service Advisor's point of view during the interviews, drivers play a crucial role in validating information and providing additional details such as visual assessments, road conditions, and load specifics. One respondent noted, "The interview about unusual sounds normally takes most of the time." Another emphasized the urgency in certain situations, stating, "If the load is medical goods, then the situation must be taken care of ASAP." However, drivers could sometimes provide misleading information, which motivates the design of the new fault-handling system. For example, "A system upgrade [on the vehicle] could trigger a new sound which makes the driver think there is a fault," said a Technician, leading to unnecessary concerns. Additionally, the truck is not fault-free sometimes and is used in a risky way, which is not ideal from safety and uptime perspectives, "Some drivers drive with fault code(s) until the engine completely breaks down, which causes more time and money to fix the truck."

4.1.2 Vision toward the driverless future

Two roles are involved in the semi-structured interviews: the Service Advisor and the Technician. The Service Advisor quickly suggests necessary actions to react to the fault based on the information collected from the drivers and SMT, and sometimes they ask the Technicians when making suggestions. The Technician is responsible for the maintenance and repair of the truck, bringing in more technical expertise. The Service Advisor role could be trained with the required competence in the future fault-handling system since they address faults when the driver cannot resolve them, and they have the prior knowledge and experience of using remote data reading tools.

During the field visits and semi-structured interviews, respondents exhibited a scared and skeptical attitude towards a driverless future, which was anticipated to be primarily due to a lack of vision and understanding about managing this transition. For instance, one of the technicians said, "There must be people behind this, " reflecting their belief that the system will be entirely non-human—specific concerns highlighted in the interviews. One of the Service Advisors noted that "drivers know the truck the best and can notice the small details," "I might be replaced by this tool," and "Communicating with driver is time-consuming … but that was also on the fun part of the job", fears of losing their jobs, anxiety about socially relinquishing control to an autonomous system, and concerns about losing the enjoyable aspects of human contact.

The interviews revealed a tone of skepticism towards autonomous solutions. However, users fall at opposite ends of the acceptance spectrum toward autonomous systems. Some interviewees recognized the potential for high efficiency and accuracy that autonomous vehicles and the new fault-handling system can offer. In contrast, those who frequently experienced mechanical faults and failures of remote reading systems tend to adopt a more conservative attitude. See Figure 13.

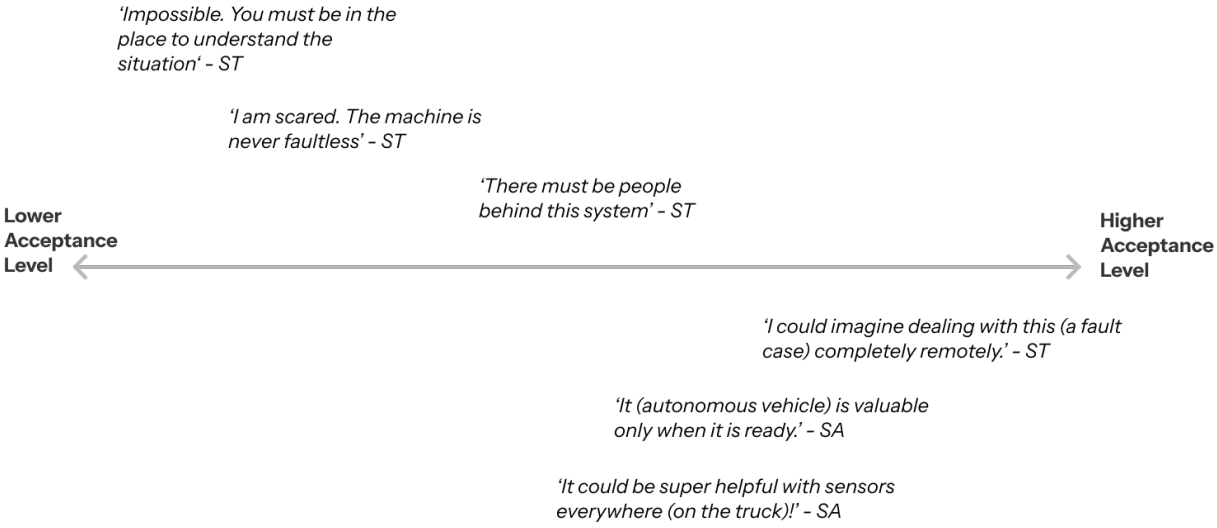


Figure 13. Polarized acceptance level toward the autonomous future

4.1.3 Enablers and Obstacles of Trust

Building trust when collecting information and making suggestions to react to faults is an important prerequisite when redesigning a safe and efficient autonomous system (Calvert et al.,

2023; Rylander & Englund, 2024). The interviews with Service Advisors indicate that trust is built by interacting with other roles with more technical expertise, and confidence is more about personal experience; one Service Advisor said, "I always make sure that I ask the most senior technician to help me with the decision (when I need a quick fault conclusion)." "Most of the time, I conclude the fault and start the preparation before the truck arrives at the workshop," said a Service Advisor with experience as a technician. By understanding the barriers and enablers of these two themes, we could reveal the principle of designing an interface that could present the data in a way that people trust and enable the users to make decisions confidently. Such reliance on human experience and knowledge and the awareness that a knowledgeable person is behind the decisions helps build trust in the system.

Several barriers contribute to a loss of trust in using remote data reading tools. Firstly, SMT only presents symptoms rather than possible diagnostics, which means the Service Advisor (SA) still needs technical support and driver interviews to make informed advice. Additionally, there was skepticism among users based on their prior experience with mechanical parts, as machinery frequently encounters problems. The perceived loss of control over the situation also plays a significant role; as one Technician remarked, "You must be in the place to understand the situation," highlighting the lack of situational awareness and control lost in remote diagnostics from the opposite perspective. Moreover, previous technical issues with network reliability and inaccurate data further compound this mistrust: "The information of fault code can sometimes be misleading," and "I always double-check the data from SMT with the driver," said Service Advisors.

4.2 Propose a future System Design

This section proposes a system design in a Transport as a Service (TaaS) context by conceptualizing the proposal system with system mapping, workflow, and interface prototypes, which are inspired by the IVMH framework (Dibsdale, 2011) and previous studies (Rylander & Englund, 2024; Tao et al., 2024)

4.2.1 Two main roles in the proposed system

Inspired by Rylander and Englund (2024), mentioned in Chapter 2, a future fault-handling system is proposed as a centralized service system separated from the service center, with a focus on two primary roles: *the Control Center Operator (CCO)* and *the Diagnosis Operator (DO)*. The *CCO* is responsible for mission delivery and fleet planning. The *DO* provides fault case analysis and suggestions on fault handling action in an advisory role to the CCO.

When the autonomous truck experiences a fault, it will make the initial decision with the alarm system and make the initial decision to react to the fault. The information on the fault and the mission status will be updated to the CCO. Intelligent tools are used here to prioritize the urgency of the fault case and make an initial diagnosis, which will be discussed in Session 4.2.2 in the future. The DO will analyze the cases and make advice, considering the competencies of the repair and maintenance services, such as the Service Center. See Figure 14.

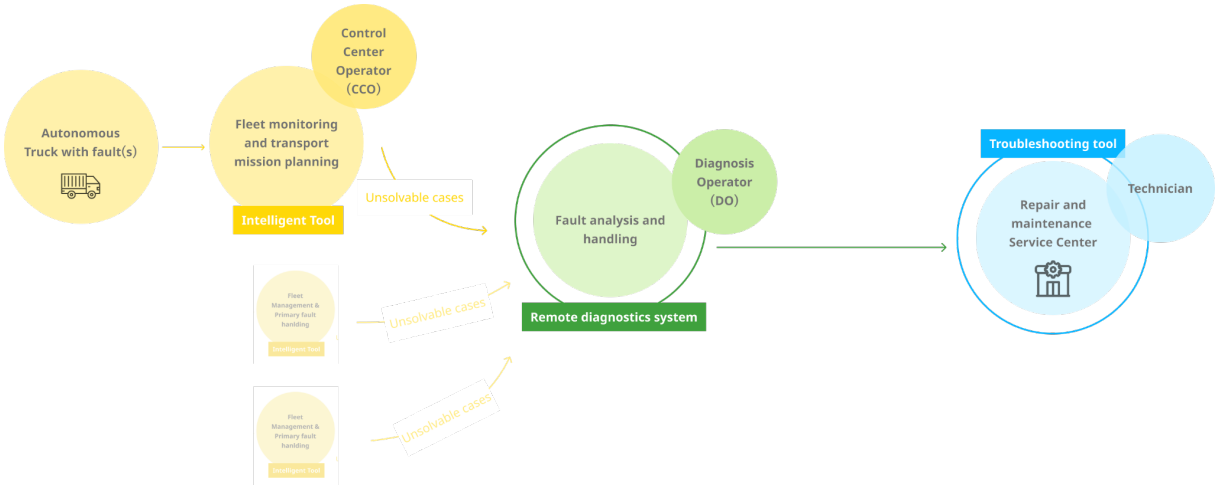


Figure 14. The proposed system mapping (Rylander & Englund, 2024)

While these roles are distinct, their physical location should be carefully considered (Dibsdale, 2011). I propose situating CCO and DO in the same location or with high connectivity, which is motivated by the current interaction between Service Advisors and Service Technicians sitting together in the service center. Furthermore, co-locating these roles would enhance situational awareness and facilitate better coordination since the decision regarding fault handling should be made from various perspectives. Moreover, centralizing multidisciplinary teams increases understanding of tasks and reduces the likelihood of mistakes or misunderstandings (Dibsdale, 2011). Optimizing resources and enabling seamless communication between disciplines fosters timely responses and directly supports uptime.

4.2.2 Future Fault-handling Workflow

In a TaaS context, the customer uses the service rather than the assets. Compared to Figure 2, this shift of ownership from the haulage company to the service providers means that the service provider needs to take care of the fleet's maintenance and performance (Duarte et al., 2024; *TRATON – Transport-as-a-Service*, 2024). In this context, the fault handling service should support vehicle health during the whole product lifetime. A fault-handling workflow is proposed here, including fault detection, initial decision-making by the autonomous truck, fault case prioritization, case assignment to DO, fault analysis, decision-making, and monitoring. An alert case is created after the truck alarm system detects a symptom or deviation. The alarm system will make the initial decision to react to this alert. This case will be prioritized based on the safety level and impact on uptime. The case will be assigned to a Diagnosis Operator based on the prioritized information and its impact on the mission. The Diagnosis Operator will analyze the case with the support of an intelligent tool and advise on the action. The follow-up services, such as Scania Assistance (*Services for Your Scania*, n.d.), will be informed by the decision. The autonomous system and Control Center will monitor the case and alert fault(s), if any. This workflow is illustrated in Figure 15.

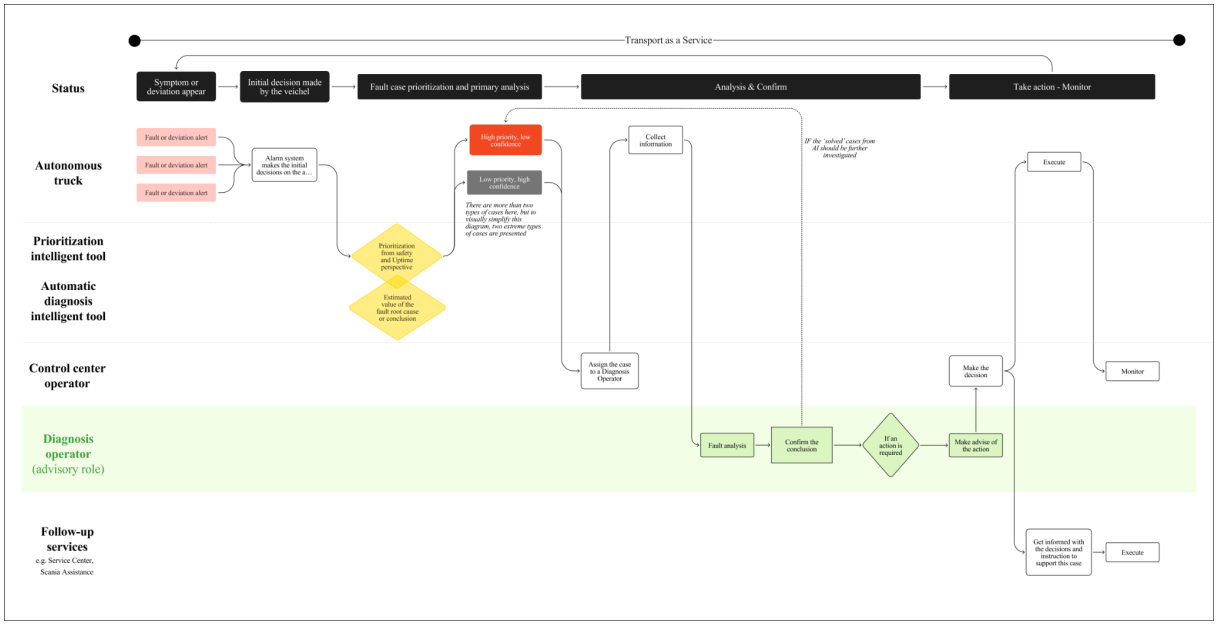


Figure 15. The proposed workflow

4.2.3 The role of intelligent tools and human actors

When designing a 'human-centric' system that prioritizes human skills, it is essential to define the critical human actors and the roles of intelligent tools (European Commission, Directorate General for Research and Innovation, 2021). Two types of Intelligent tools are proposed here to enhance workflow efficiency when handling cases, considering a team will most likely be responsible for one or more fleets.

Currently, evaluating the risk of a fault is one of the tasks of the Service Advisor when giving suggestions on fault handling. This means the Service Advisor needs to provide suggestions based on whether this fault will cause a major breakdown and impact the mission. Based on this finding, the first intelligent tool is proposed to prioritize cases from safety criticality and uptime perspectives. When faults or deviations are detected, they are categorized based on their priority and probability. High-priority issues should be addressed immediately to ensure safety and minimize downtime.

As mentioned in Session 4.1, one of the barriers to using a remote data reading tool is that the current interface needs to be more informative and present data without giving possible analysis assistance. Meanwhile, Service Advisors often look for suggestions from the Service Technicians when they are not sure about the analysis. The second tool, for automatic diagnosis and support analytic process, utilizes machine learning algorithms and historical data to analyze faults and provide estimated values and the probability of the fault's root cause. For example, it might indicate a 95% confidence that the brake pad kit is broken. For example, cases with lower confidences, those between 0-80%, will be investigated by human agents (Diagnosis Operators). The human agent will start by collecting information, conducting a thorough analysis, and making conclusions about the features provided by the digital interfaces. Cases

with higher confidence, for instance, those with more than 85%, also require confirmation from Diagnosis Operators to verify the findings and investigate further if necessary.

Human actors in this workflow are crucial for analyzing and confirming diagnostic conclusions. Their actions and data will feed feedback into the model, continuously improving the algorithms and enhancing the tool's support for human operators. Incorporating AI tools into the fault diagnosis workflow offers significant efficiency, scalability, and data-driven decision-making advantages. However, these tools have limitations, including reliance on known data, inability to change ultimate goals, and lack of contextual understanding, which align with the concepts of 'not-knowing' and 'episteme' from the Seven Elements of Phronesis in Human and Autonomous Decision-Making (Koutsikouri et al., 2023).

4.4 Prototyping

To make it more customized to each audience's needs and more efficient, the system should be built to share the same primary monitoring data and present relevant data based on the operator's identity (Few, 2013). In this case, two screen displays could fit well to share information that shares situation awareness and presents the relevant data concerning different roles. To fulfill the needs of diverse profiles of audiences and serve other purposes.

4.4.1 Screen A: Provide an Overview to Share Situation Awareness

This interface is an information dashboard to share situation awareness in an interdisciplinary team. It is designed to keep the Control Center and Diagnosis Operators aware of the situation. It displays vital information modules on a single-screen display, providing updated situational awareness, which supports vehicle management and decision-making and keeps the team working towards the same goals. This screen offers situational awareness by providing an overview of the fleet status regarding uptime, operational data, fault overview, and fault case assignment. It serves as a tool to bring two roles, the Control Center Operator and Diagnosis Operator, on the same page to make the user feel in control, foster a smooth collaboration, and support connectivity by sharing information that concerns each role. According to Few's (2013) suggestion, the screen type should be defined by size, viewing distance, ratio, and resolution. Here, I suggest the aspect ratio as 16:9 without defining other aspects, such as the screen size, since the room setting and the number of people in the exact physical location are undefined.

The prototype of Screen A is presented in Figure 16, and four sessions are labeled.

- The overview of fleet operations regarding uptime and downtime, as well as the team members that are online. (A - Overview data and team)
- Location is an important part of what's being tracked when making action decisions, considering the infrastructure and mission status (Few, 2013). Hence, a geospatial display is needed so that users can check out the mission status and infrastructures around the alert cases. (A - Map)
- Alert cases are categorized based on the status of the proceeding. Each case is labeled by the prioritization level and mission status, with critical information presented. (A - Alert cases)
- A navigation menu with archive cases, relevant people, and setting features. (Nav menu)

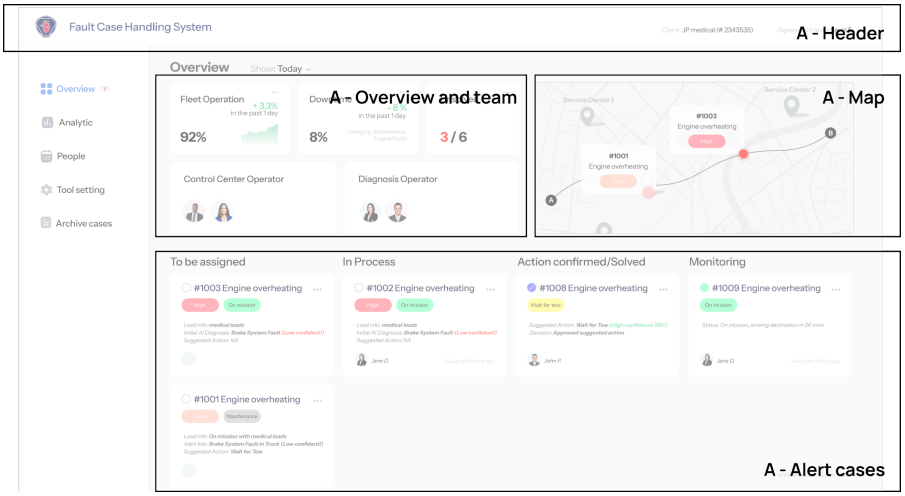


Figure 16. Session labeling of Screen A

4.4.2 Screen B: Analytic Tool for Diagnosis Operator

Screen B is an analytic tool dedicated to the Diagnosis Operator, specifically for fault analysis with detailed information. It includes detailed symptom information (Englund, 2021; Rylander & Englund, 2024) and previous communications or history related to the vehicle, ensuring operators have the full context of the issue.

- The profile of this Diagnosis Operator is shown on the top. All case(s) assigned to this Diagnosis Operator are presented in a list. (B - Case Overview)
- Basic information with vehicle information and operational data. (B - Basic Info)
- Information and tools for analyzing faults, including the criticality of the fault, symptom details, and confidence levels in AI Diagnosis. This supports operators in assessing and prioritizing issues effectively. The interface integrates an intelligent tool called 'AI Diagnosis,' which provides an initial analysis of the fault component. Operators can view the analysis process, understand the data sources, adjust parameters, and re-evaluate decisions as needed (Rylander & Englund, 2024; Tao et al., 2024). (B - Analysis)
- Features that support decision-making and facilitate coordination with external services, such as tow trucks or other service domains, streamlining communication for the next steps. The interface also has a call-to-action feature, allowing operators to immediately inform the team or escalate the issue, ensuring that the workflow remains responsive and coordinated across the team. (B - Decision)
- A navigation menu with archive cases, relevant people, and setting features. (Nav menu)

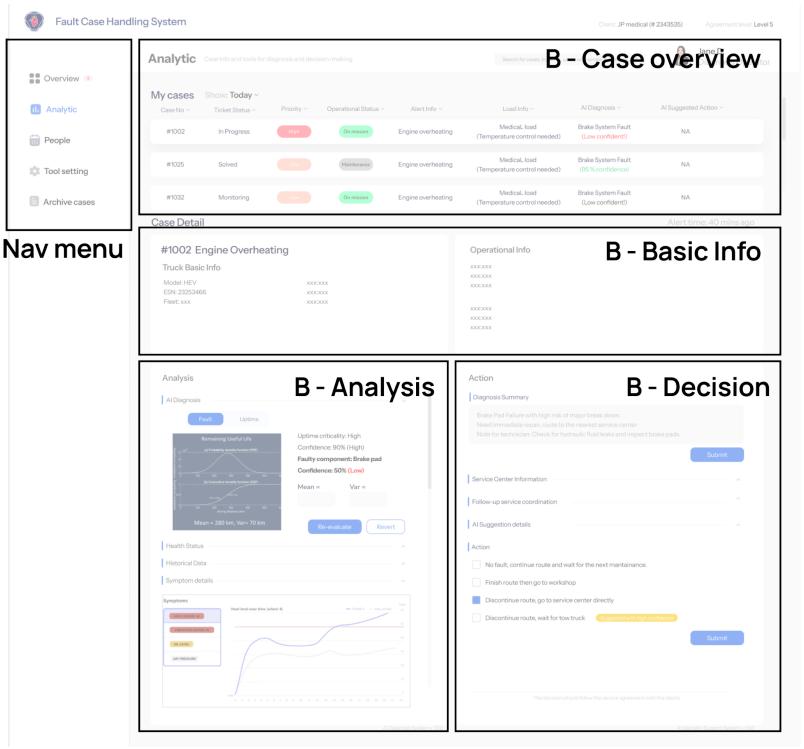


Figure 17. Session labeling of Screen B

4.5 Design Feedback and Iteration

The overall reaction to the prototypes from the Design Feedback Workshop at Scania was positive, and opportunities to improve the interfaces are discussed and summarized in Table 6. ‘There are many details on these interfaces,’ said one of the design feedback workshop participants at Scania. The feedback on Screen A is mostly about the map feature and information about the alert case card; see Feedback A01 and A03. The group is primarily satisfied with Screen B, which includes discussions regarding communication features like Feedback B20.

Table 6. Summary of the opportunities for improvement

Interface session	Feedbacks	Improvement
A - Overview data and map	A01. The map feature is essential in this interface to visualize the mission status. Currently, the map only presents the alert cases, which means it will be empty when there is no alert case. It's suggested that a map presenting all online vehicles should be provided to provide a sense of control and increase system transparency.	A02. A bigger map with tabs presenting vehicles of different statuses.
A - Header	A10. The interface should consider situations where more than one fleet is under monitoring.	A11. The default mode would present cases from all the fleets, with filter features and clear case labeling.
A - Alert cases	A20. The information on each card could be simplified with the most essential information.	A21. Simplify the card information and reduce the card size to create space for adjustments to the map feature in A11.
B - Analysis	B10. The fault code information is missing.	B11. Provide contextualized information on the fault codes
Nav menu	B20. Features that support communication between the team and other disciplines are suggested to enhance connectivity and traceability.	B21. Add a tab called 'Scania Assistance' to emphasize the communication with the follow-up services.

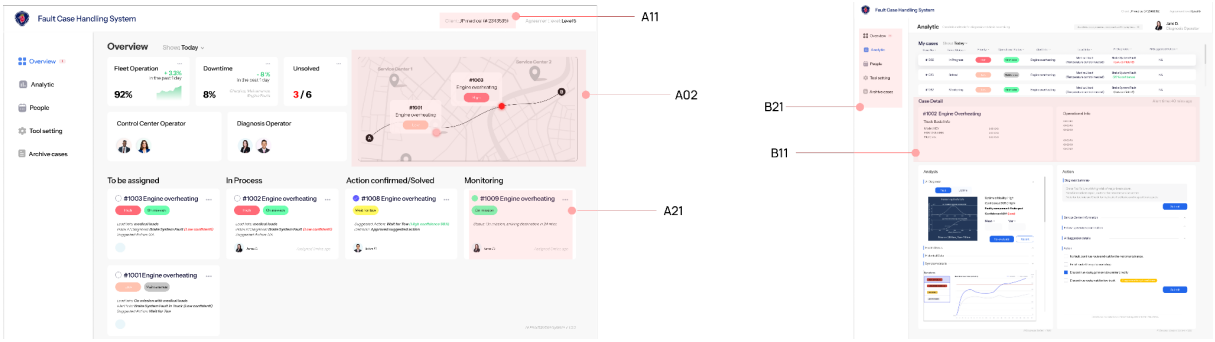


Figure 18. Opportunities of improvement

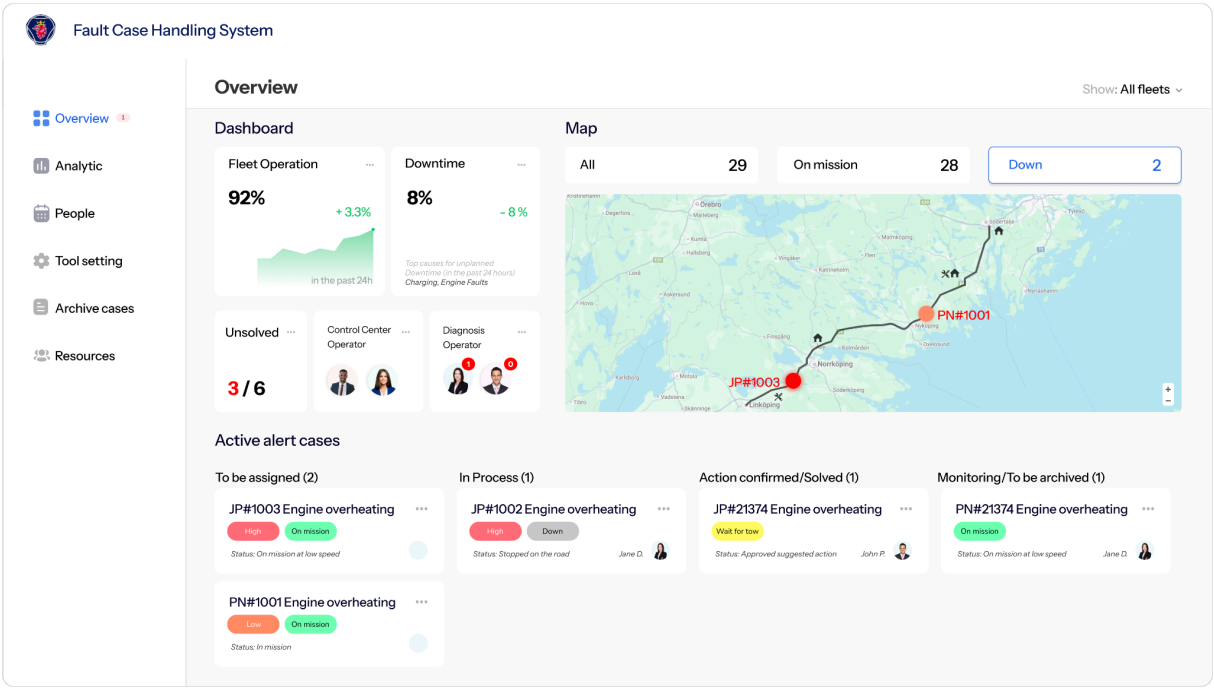


Figure 19. Finalized prototype - Screen A

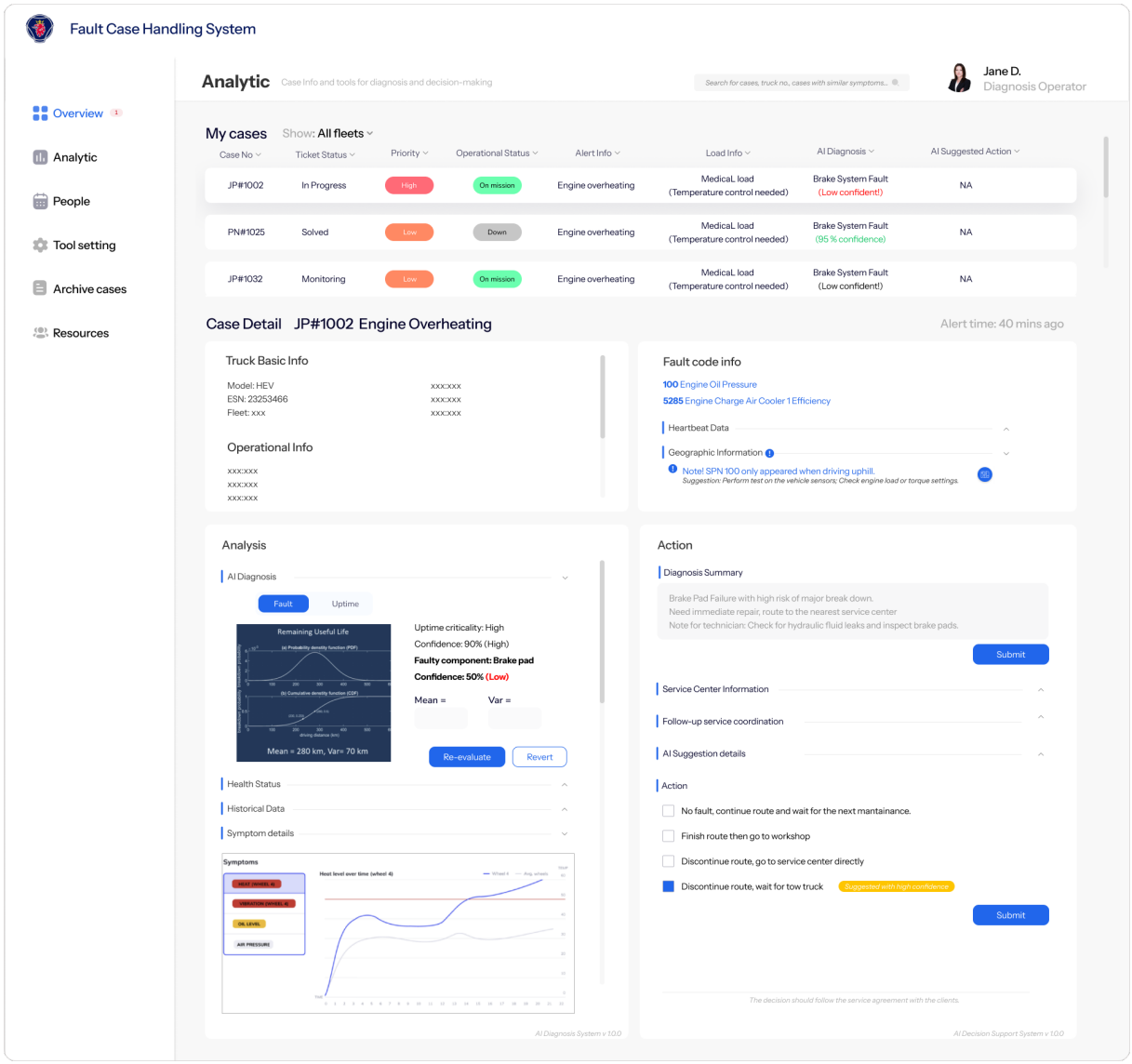


Figure 20. Finalized prototype - Screen B

5 DISCUSSION AND CONCLUSION

This chapter discusses the results from Chapter 4 and concludes the findings.

5.1 Design Considerations

Trust building and usability are paramount in designing the human-machine interface (HMI) for a fault-handling system in driverless vehicles. Inspired by Dibsdale's (2011), Calvert et al.'s (2023), Rylander and Englund's (2024) works, and user studies, this session elaborates on the findings and proposes design considerations for the future fault-handling system. These considerations may overlap as they work synergistically to deliver a subjectively trusted and user-friendly system, defining the user's expertise as the Diagnosis Operator. As mentioned, 'trust' has been identified as one of the key factors in the transformation to a driverless future. In this case, it means the trust between the service providers and the fault-handling system.

5.1.1 Tracing Condition: Defining human agents

Currently, Service Advisors sometimes consult Service Technicians when they need a quick and accurate analysis of the fault. It reflects the reliance on the human experience of knowing a knowledgeable person is behind the decision. Tracing conditions stress the importance of identifying human agents who understand the system's capabilities and roles and recognize the moral consequences of the system's behavior (Calvert et al., 2023). In fault handling, this means ensuring that the roles of key human agents, such as Diagnosis Operator and Control Center Operator, are clearly defined and that these individuals are knowledgeable about the system's operations. This ensures that even without a driver, control can be traced back to competent human agents who oversee and manage the system's response to faults. This traceability fosters accountability and trust, ensuring the system operates efficiently and ethically, meeting human-centered control conditions, as Matthias (2004) described.

5.1.2 Situational awareness at a glance

Providing situational awareness at a glance is crucial for users to feel in control since it addresses the concerns of losing control of the situation when no human actor is physically present with the vehicle (Calvert et al., 2023; Dibsdale, 2011). The system should provide a quick and comprehensive overview of critical data, enabling rapid appraisal and readiness for any unit member, which enhances situational awareness and supports the tracing condition. For example, as stated in Chapter 4, in a control center setting, the interface(s) could display an overview of fleet status regarding assignments and vehicle health.

As Calvert et al. (2011) argued, From an individual user's perspective, this principle allows users to grasp the current situation quickly, especially when they have been "out of the loop." This requires understanding what information is crucial for each role and the whole team to be aware of the situation. From a team collaboration perspective, it can enhance the situational

awareness of human agents working in the same workflow (Dibsdale, 2011), thereby improving collaboration and supporting connectivity, which will be discussed later.

5.1.3 Transparency

Transparency in both the data itself and the automation process is essential. Trust is closely linked with understanding the provenance of data and the reasons behind automated decisions. A system that allows users to understand where the self-diagnosis might have gone wrong and the rationale behind those decisions can significantly enhance trust. Dibsdale (2011) suggested that provenance could be assured with data transmission assurance, certificates, and transactional integrity.

As Xin et al. (2024) discussed, one way to avoid the 'black box' perception is to present the information used by the intelligent tools and the decision-making process on the interfaces, such as safety criticality, faulty diagnosis, fault prognosis results, and confidence. Such information should be integrated during the analysis process to ensure a smooth user flow when the Diagnosis Operator analyzes the case and confirms the decision. Furthermore, interaction between users and intelligent tools should be enabled by adjusting the decision-making model with the information in which they trust. Such adjustment also enables humans to explore different aspects of the situation rather than use only known data, which refers to 'Not-knowing' and 'Lived experience' as the advantages of human judgment (Koutsikouri et al., 2023).

Meanwhile, transparency in the analyzing process helps to provide objective information and avoid biases—the 'black box' effect in human decision-making (Bonezzi et al., 2022). In this context, enhancing the transparency of the decision-making process is beneficial when avoiding the black boxes of both human and algorithmic decision-making (Bonezzi et al., 2022).

5.1.4 Connectivity

As mentioned in Chapter 4, trust is built by interacting with other roles in the current system. The system should provide features and information supporting smooth interaction between actors. This involves creating features that facilitate seamless communication among various stakeholders within the system. Enhancing connectivity ensures the system supports collaborative fault handling and improves overall system efficiency.

5.1.5 Tracking Condition: Situational responsiveness

The tracking condition considers the responsiveness of a system's behavior to human (moral) reasons and intentions to act (Calvert et al., 2023). It emphasizes the need for automated systems to be responsive to human (moral) reasons and intentions, ensuring that the system's behavior aligns with human values and situational requirements. In the context of fault handling, the system should be capable of recognizing and responding to both specific personal needs (proximal) and broader societal values (distal). For example, the system should react differently to monitoring and fault diagnosis according to the use goal and service agreement between the service provider and customers.

5.1.6 Usability

In application software design, achieving effective usability and data presentation can support the diverse needs of users. Dibsdale (2011) highlights that users vary widely in experience, skills, and education, which affects how they interact with software. For instance, senior managers may need high-level summaries, while data clerks require detailed data. The software should offer customizable interfaces that cater to different user roles and experience levels. Meanwhile, it should support various usage patterns, from ad hoc interactions requiring quick access to information to continuous use where advanced features and customizations are beneficial.

Furthermore, the software must handle diverse data types and presentation needs. This includes supporting qualitative and quantitative data, historical records, and free-form information. Effective design involves providing flexible data views and tools that allow users to access and analyze information according to their needs. By incorporating these considerations, the software can offer a more tailored and intuitive user experience, enhancing usability and data management.

5.1.7 Competencies of the Diagnosis Operator

With a focus on designing interfaces for Diagnosis Operators for fault-handling, one of the goals of this thesis is to identify the potential users who will take care of fault handling, specifically fault analysis, and support decision-making. It is suggested that in the current system, Service Advisors with technical backgrounds or knowledge have the potential to play the future role of remote Diagnosis Operators. As a Diagnosis Operator, making decisions for efficiency and safety is the top priority, which aligns more closely with the current role of Service Advisors than Service Technicians. Meanwhile, Service Advisors already handle faults remotely, which is an integral part of their role. Few (2013) discussed an example in the airline industry where pilots are trained to periodically scan cockpit displays, highlighting the importance of regular training on specific information and interface familiarity. Therefore, it is recommended that training be conducted to facilitate the transition of Service Advisors into remote Diagnosis Operators. The training should cover intelligent tools, the technical knowledge required for electric vehicles (EVs), and the effective use of interfaces. Service Advisors can efficiently and safely manage fault-handling by equipping them with these skills in a remote diagnosis context.

5.2 Design for a Speculative Field

5.2.1 Service Dominante Logic

The shift from a product-centric business model (G-D Logic) to a service-centric one (S-D Logic) signifies that the value conveyed is no longer tied to the product but to the integrated solutions encompassing tangible and intangible assets. The value is delivered from a service package in the context of transport as a service (TaaS), requiring network structure and technology adjustments since the current fault-handling service is more of an ‘extra’ part (Tukker, 2004) instead of the center of the business.

Furthermore, removing the driver from the system means that the system must undergo systematic changes to compensate for the absence of this key actor. This study further validated the driver's role in diagnostics, while human-caused downtime was also identified. For example, locating the source of sounds or vibrations is time-consuming since it depends on whether the driver is familiar with the truck, or sometimes drivers are not proactive in solving the faults if the trucks can still drive, etc. These findings highlight the motivation to develop a more efficient fault handling for autonomous vehicles, enhancing system efficiency and fostering a more sustainable transformation.

In the current fault-handling system for driverless vehicles, the relationships and interactions between various roles reflect the invisible social structure, which is the central design material in Service Design (Vink, Koskela-Huotari et al., 2021). These roles determine how responsibilities are divided, decisions are made, and information flows, all influenced by the broader organizational framework and norms. Thus, they inspired the design of the future fault-handling system. For instance, the distribution of authority and the division of labor can affect how new technology, such as AI prioritization and diagnostic tools, is adopted to prioritize and assign fault cases. Meanwhile, Wilmering and Davies (2011) note that introducing new systems can lead to unintended changes in behavior, highlighting the need for understanding the underlying social dynamics. The finding that trust can be built by interacting with other roles and accountability of the action has informed the development of design considerations such as tracing conditions and connectivity.

5.2.2 Research Methods

One key aspect encountered during this project is the challenge of applying human-centered design principles when users are not yet involved. Vink et al. (2017) emphasize that aesthetic disruption is vital in designerly practices for revealing and altering these mental models. Conducting field visits and semi-structured interviews helped identify the concerns and barriers that potentially hinder a driverless future, even though the interviewees are not the real users of the future system.

I also recognized the difficulties of involving potential users in the design process during the semi-structured interviews when the storyboards were presented and discussed. As mentioned in the Result Chapter, polarized attitudes toward the driverless future exist. When system-level changes involve transformation, actors often fall into "defensive routines" that protect their established mental models from critical examination (Vink et al., 2017). This highlights the need for training and habit-building around digital tools to collect information traditionally gathered from drivers (Few, 2013).

Throughout the two workshops, discussions about business models were essential for interpreting feedback and further developing the proposal, grasping the diverse perspectives of different departments within the organization regarding the vision for the future fault-handling system. The first workshop, in particular, discussed service agreements and the extent of responsibility service providers hold for vehicle maintenance. For example, higher service agreement levels grant the company's service team greater authority in fault handling. This insight inspired me to adapt the proposal to align with the Transportation as a Service (TaaS) concept, incorporating business model considerations directly into the workflow design.

In this study, Copilot generated storyboards for semi-structured interviews. Using algorithms in the design process helped accelerate the workflow, allowing the focus to remain on idea generation and refinement through brainstorming and prompt clarification.

5.2.3 Design artifacts

As mentioned in Chapter 3, this project used the strength of design to explore the possible future of fault-handling through making and critiquing artifacts that function as proposal solutions (Zimmerman & Forlizzi, 2014). The design artifacts used in this project are in various forms for different purposes in each stage.

The storyboards, as mentioned in Chapter 3, were generated from Copliot. It was used as a tool during the semi-structured interviews to visualize the possible future of fault handling, engage interviewees' senses, and trigger discussion on aspects that might not be covered when words purely describe the scenarios. Meanwhile, interviewees had different interpretations of the visuals, which shows their distinct focus on the system and reveals their mental models. For example, most Service Technicians prefer the scenario in the storyboard where the user could feel the simulated symptoms, aligning with their role in the current system, such as future diagnosis and repairing the truck by conducting test driving and experiencing the faults.

System mapping was developed and used after the semi-structured interviews, visualizing the complex system by highlighting key actors. It's impossible to know all the parts of a system to be accountable for a complex system (Sevaldson, 2022). Instead of figuring out all its details and interconnectedness, system mapping provides an overview of the system by mapping out the crucial parts of the fault-handling process. This method was used to map out the current system and the proposed system with the main roles and how the interaction between these roles supports the fault-handling process, providing a clear picture of the current system and aligning our understanding of the system. Meanwhile, these system mappings triggered the discussion beyond the mapping scope, such as service networks and infrastructures that support the mission and warranty, which are regarded as crucial when handling faults in the future and considered in the workflow design. The key takeaway is that effectively visualizing essential actors and the overall structure doesn't require listing every detail. By inviting diverse perspectives into the discussion, any missing key actors or interactions are likely to be identified, which highlights their importance. This approach ensures that critical elements are brought into focus naturally through collaboration.

The workshops show that visualization is particularly valuable in communicating proposals for change, which is the change of value creation in this context (Sevaldson, 2022; Vink, Koskela-Huotari, et al., 2021). Currently, communication between drivers and Service Advisors creates a part of the value. Removing drivers from this system means a change in value creation with a shift to technology-driven remote diagnosis, which was visually presented through the system mapping. The System mappings presented such change and helped participants to provide concrete and specific opinions. For example, during discussions, respondents would refer to specific elements by saying phrases like "the yellow bubble" or "the blue bubble," which fostered informative and easy-to-follow discussions.

The proposal sparked discussions and redefined what the fault-handling system could become. Prototype presentations provided a tangible medium for cross-departmental discussions, serving as a common language for the group to engage in dialogue—whether about the prototype itself or the underlying business model and technical capabilities.

5.3 Conclusion

Returning to the research question: ‘How can we enhance the efficiency of fault handling and build trust when interacting in a remote setting for driverless trucks?’ This thesis explored the challenges of fault-handling in autonomous vehicles used in hub-to-hub operations, with two main purposes: to navigate the challenges and opportunities in this transformation regarding the relationship between humans, remote interactions, and intelligent tools by understanding the current fault-handling system, to propose and prototype a system design concept, developing knowledge on presenting information and supporting decision-making for remote fault handling. The purposes are achieved, which deepened the understanding of the current system and developed knowledge and artifacts that could benefit future research.

The fieldwork highlighted a key leverage point when building trust: the relationship between humans and technology, providing the user perspectives regarding the future fault-handling system. The insights inspired the conceptualization of a remote fault-handling system. The proposed design and interaction concept aims to provide a tangible and visualized vision to spark discussion within the organization. The proposal, with system mapping, workflow, and interface prototypes, conceptualized the possible future of fault handling and further developed the findings from previous research. It demonstrates how AI tools can automate fault diagnosis and decision-making by integrating AI tools into the fault-handling workflow.

Meanwhile, the study emphasizes that defining the future business model of fault handling is crucial for creating a more realistic design concept. Interactions between service providers, customers, and technology must align with business goals. Currently, services for vehicle health are provided as separate parts, and clients order the services with a certain service scope. For example, higher service agreements grant service teams greater authority in managing faults, influencing how fault-handling solutions are delivered. In a TaaS context, adapting or changing the business model into the driverless transport system is necessary since the service scope will be extensive. These services need to be integrated to support vehicle health by identifying their roles in fault handling for the product's lifetime.

Visual prototypes and system mapping throughout the project proved valuable in making complex ideas accessible. During two workshops at Scania CV, design concepts were discussed from both business and human-machine interaction perspectives. These discussions underscore the benefits of leveraging design to foster collaboration and drive meaningful conversations about system transformations and future challenges.

6 LIMITATIONS AND FUTURE RESEARCH

This chapter discusses the limitations of this thesis and the opportunities to bring the research forward.

6.1 Limitations

This research primarily relied on qualitative methods, including workshops and semi-structured interviews, which provided in-depth insights into the fault-handling system. However, the reliance on qualitative data may limit the generalizability of the findings. The specific context of the case company and the small sample size make it difficult to apply the conclusions across other organizations or industries. Future research could incorporate quantitative methods or broader sampling to validate and expand upon the results.

A key limitation of this study is the incomplete definition of how external functions interact with the proposed fault-handling system. While the internal system design was mapped thoroughly, the integration of external actors—such as service centers, customers, or cargo receivers—was not fully explored. This gap in the workflow design was partly due to the project's limited scope. Future research should focus on refining these interactions, ensuring they are clearly outlined within a comprehensive system framework to improve overall efficiency and communication in fault-handling processes.

Another limitation is the lack of early-stage collaboration with key technical roles such as AI engineers and data scientists. Their involvement in the system planning process could have enriched the integration of AI into the fault-handling system design. AI-driven solutions are critical for automating diagnostic processes and improving decision-making, and the absence of these competencies limits the potential for innovative service design. Future projects should prioritize incorporating AI expertise from the outset to fully explore the potential of AI in enhancing system efficiency and user interaction.

6.2 Future Research

Future research is recommended from both business and user experience perspectives.

From a business standpoint, further investigation is needed to explore how the business model in a TaaS impacts the fault-handling system design and to determine if specific service scopes are required as the business transitions into a more service-oriented phase, particularly when customer ownership shifts. In the current system, service agreements set rules and procedures, defining acceptable behaviors, service scopes, and limitations, such as remote access to vehicle data. These agreements also determine the responsibilities of Service Advisors when accessing data remotely and providing fault-handling advice.

On the user experience side, future prototype development should address specific needs in defined scenarios. Conducting user research with vehicle managers and Service Advisors can help identify critical sub-scenarios for each screen. For instance, in Screen A, active alerts are presented as cards with prioritization and mission status conveyed through color and text. Exploring whether this information is sufficient for fleet managers or future Control Center Operators to assign cases will be essential.

Information hierarchy also needs further consideration to align with real-world cases. In the design feedback workshop, we discussed whether each alert card should represent an individual fault or an entire vehicle, especially if multiple faults are interrelated. If faults are independent, each card should represent a single fault. If interconnected, the card should represent the vehicle. Another key discussion revolved around whether a risk tolerance range should be defined. This challenges the assumption that trucks must be fault-free and opens avenues for defining what constitutes a "safe truck" within acceptable risk levels.

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APPENDIX

Discussion guide InterviewLength: 30-45 min	**Service advisors**	**Service Technician**	**Trigger materials**
2min Background & Warm-up - Name, role - How long have you been working in this role - Set the scene: Start talking about driverless context and autonomous trucks	x	x	Pictures of driverless vehicle
5 min Current* Challenge of remote diagnosis - What tool are you using for remote diagnosis? - Does it work with other tools like plug-in tools? - What's the **challenge** of doing the diagnosis remotely? (Trust?) - Screenshot of the interface or take photos of the dashboard or key screens	x *Assumption:* *People have different ways to get into a fault and solve it.* *People have different acceptance levels of remote diagnosis/technology.* ***Observe how confident they are in their ability to form hypotheses remotely, which is highly related to***		

	their acceptance of remote tools.		
5 min Current* Observation/scenario role-play - Choose one of the scenarios. - How would you solve the situation with the current tool?	x	X Is there any information that is often missed from the drivers and the digital tools that you need to get yourself?	Scenario description
10 min Future* In the cases that you describe, if the driver is not responding or the truck is driverless - What do you expect in this driverless context? How would you assume it to work? - I have a list of mechanical symptoms here. Show the list of symptoms: What are the most important symptoms? What are the **challenges** and opportunities for remote communication when the driver is absent?	x		-symptom list -post-it
10 mins Future* Stimuli Stimuli 1: Concept description about remote operation, which is a simulation of the driving experience	x	x	Storyboards

Stimuli 2: Articulate the data and present it on the dashboard (guide the direction of discussion and make them understand what the idea is to avoid fixation) *Assumption: Stimuli 2 fits with the current mental model more since they are collecting information on symptoms from the drivers instead of experiencing symptoms themselves*			